Full Paper Submission to CI26 (2025), **xx**:xx 1–16

CAMBRIDGE
UNIVERSITY PRESS

APPLICATION PAPER

Dissipating the correlation smokescreen: Causal decomposition of the radiative effects of biomass burning aerosols over the South-East Atlantic

Emilie Fons[1], Isabel L. McCoy[2,3,4], Tom Beucler[5], David Neubauer[1,6] and Ulrike Lohmann[1]

[1]Institute for Atmospheric and Climate Science, ETH Zurich, Zurich, Switzerland.
[2]Cooperative Institute for Research in Environmental Sciences, University of Colorado, Boulder,Colorado, USA.
[3]Chemical Sciences Laboratory, NOAA, Boulder,Colorado, USA.
[4]Now at: Department of Atmospheric Science, Colorado State University, Fort Collins,Colorado, USA.
[5]Faculty of Geosciences and Environment, University of Lausanne, Lausanne, Switzerland.
[6]Now at: Urban Climate, GeoSphere Austria, Vienna, Austria.

Keywords: causal inference, biomass burning aerosols, smoke, marine low clouds, earth observations

Abstract

Biomass burning aerosols (BBAs) from Southern Africa seasonally overlie the semi-permanent South-East Atlantic (SEA) stratocumulus deck, impacting the region's energy budget through complex aerosol-cloud-radiation-meteorology interactions. Climate model intercomparison initiatives, like the Aerosol Comparisons between Observations and Models (AeroCom), have highlighted the large inter-model variability for BBA radiative effects, especially over the SEA, due to parameterization of emission modeling and smoke properties. Observational constraints are needed to reduce these uncertainties, but correlative observational studies are typically affected by confounding meteorological influences. We propose a physically informed statistical approach, based on causal graphs applied to satellite observations, to disentangle BBA influences on shortwave radiation over the SEA and identify the main sources of statistical biases plaguing observational studies. We find that, during the fire season, BBAs cause a regional shortwave cooling of -2.5 W m^{-2}, which can be decomposed into equal contributions from three physical pathways: aerosol-radiation interactions (ARI), adjustments to ARI, and aerosol-cloud interactions (ACI). We also perform ablation experiments with graph variants to investigate the main sources of confounding — like large-scale winds, humidity-biased retrievals or spatial aggregation of data — and show that they result in biased radiative effect estimates (-50 % and +15 %). Once free of such biases, our derived causal estimates of smoke radiative effects can be used as observational constraints to improve climate models.

Impact Statement

We apply a rigorous causal inference approach to identify and remove biases impeding the quantification of the radiative effect of smoke aerosols from earth observations over the South-East Atlantic, a region severely affected by biomass burning aerosols five months a year. Such a bias-free observational estimate can be used to constrain climate models struggling with the correct simulation of smoke-cloud-radiation interactions and of their impacts on climate change and precipitation extrema.

1. Introduction

Not only are biomass burning aerosols (BBAs) deleterious for air quality, they also modify the radiative budget of the Earth (Reid et al 2016; Szopa et al 2021). Smoke radiative effects obtained from climate

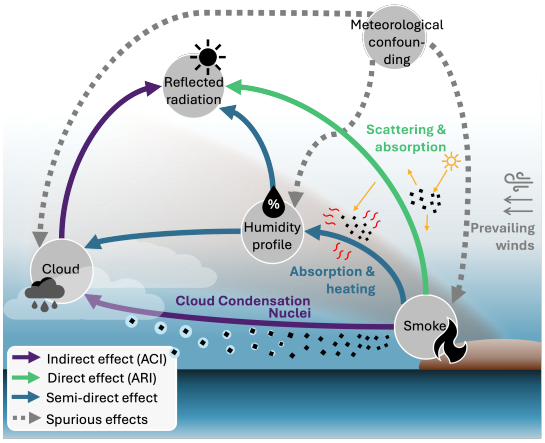

Figure 1. *Schematic causal pathways for the shortwave radiative effects of BBAs. The direct, indirect and semi-direct pathways are shown in green, purple and blue, respectively. In grey, dotted arrows show how confounding meteorological influences can bias the statistical estimate of BBA radiative effects.*

models (e.g. AeroCom II models) suffer from uncertainties related to fire emission modeling and parameterization choices (e.g., Myhre et al 2013; Zhong et al 2022). With increasing risks of wildfires due to global warming (Parmesan et al 2022), it is essential to constrain the uncertainties surrounding BBA radiative effects.

Due to the complexity of smoke-cloud-radiation-meteorology interactions, reducing this uncertainty is not trivial. One third of global BBA emissions originate from fires in Southern Africa during the biomass burning season (van Marle et al 2017). Large-scale winds blow the smoke from Southern Africa towards the marine boundary layer (BL) clouds of the South-East Atlantic (SEA). BBAs are composed mostly of Black Carbon (BC) and Organic Carbon (OC), with different radiative properties: BC strongly absorbs shortwave (SW) radiation, OC mostly scatters SW (Reid et al 2005). Thus the sign of BBA radiative effects dependent on BBA composition, but also on the underlying surface: BBAs can have a SW cooling effect over dark surfaces like the ocean but a warming effect over bright surfaces like stratocumulus decks. Additionally, freshly emitted BC and OC are hydrophobic, but they can age during atmospheric transport, become hydrophilic and eventually serve as cloud condensation nuclei (CCN) for cloud formation (Reid et al 2005). Fig. 1 summarizes the complexity of the three pathways by which BBAs interact with radiation: (1) the direct effect, also termed aerosol-radiation interactions (ARI), corresponding to the direct absorption and scattering of SW by aerosols; (2) the indirect effect, also termed aerosol-cloud interactions (ACI), describing how hygroscopic aerosols can serve as CCN and modify the radiative properties of clouds (rapid adjustment to ACI); (3) the semi-direct effect, considered a rapid adjustment to ARI, whereby aerosol absorption of SW radiation modifies the atmospheric profiles of temperature and humidity, thus impacting cloud formation (Bellouin et al 2020). Even though smoke can also interact directly with longwave (LW) radiation, the direct SW effect will dominate (e.g., Stone et al 2011). Besides, SEA clouds will have a negligible LW effect at the top of the atmosphere due to their low altitude (Lohmann et al 2016b), hence the focus on SW radiation in this study.

Climate models can be used to compute and decompose aerosol radiative effects as the difference between a control and a counterfactual scenario. Examples of counterfactual scenarios include switching off all anthropogenic aerosols emissions (Ghan 2013) or switching off the ARI pathway by modifying a model's radiation or cloud scheme (Diamond et al 2022). However, individual parametrization choices lead to high inter-model variability of aerosol radiative forcing estimates, especially over the SEA (Myhre et al 2013; Stier et al 2013). Estimating radiative forcing from observations is not so straightforward,

as co-variability of large-scale meteorological conditions induces confounding that biases the aerosol-cloud-radiation relationships (Fig. 1).

In this study, we use causal inference (Imbens and Rubin 2015; Pearl 2009) to compute BBA radiative forcing from 16 years of satellite observations and reanalysis data. We use the causal framework to rigorously justify the choice of meteorological control variables, allowing us to decompose the radiative forcing into its direct, indirect and semi-direct components, and to identify and estimate confounding biases originating from meteorological influences and methodological choices.

2. Data and Methods

2.1. Satellite and Reanalysis Data

The dataset comprises 16 years (2003 - 2018) of cloud, aerosol and meteorology data for the South-East Atlantic (SEA) low-level cloud region, which is impacted by smoke during the biomass burning season (30°W-15°E- 25°S-0°, June-October).

The cloud observations are obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS, Platnick et al 2015) and the Clouds and the Earth's Radiant Energy System (CERES, Doelling et al 2013, 2016) instrument aboard the Aqua satellite. The meteorological variables are downloaded from the ERA5 (Hersbach et al 2018a, 2018b) and CAMS reanalyses (Inness et al 2019) from ECMWF. The products used in this study are: daytime cloud fraction (CF), cloud optical depth (COD), cloud droplet effective radius (r_{eff}) and Aerosol Optical Depth (AOD) from the daily level 3 $1° \times 1°$ MODIS product; the reflected top-of-atmosphere (TOA) shortwave flux (SW $\uparrow$) from the daily level 3 $1° \times 1°$ CERES product; pressure-level wind speeds and relative humidity from ERA5; pressure-level mixing ratios of BC and OC from CAMS. Using the adiabaticity assumption (Brenguier et al 2000; Quaas et al 2006), we derive the cloud depth H from COD and r_{eff}. The key variables are listed in Table 1.

The data are co-located on the $1° \times 1°$ grid of the MODIS data using a conservative regridding approach (Zhuang et al 2023) and the reanalysis data are interpolated to the approximate local crossing time of the Aqua satellite. Among cloudy pixels, only low marine cloud pixels, identified as having a cloud top pressure larger than 680 hPa, are kept in the analysis.

Table 1. *Causal graph node variables and their descriptions.*

Graph node	Description
BBA	Biomass burning aerosols. By default, this is CAMS' total mass of BC and OC in the atmospheric column. Replaced by MODIS AOD to evaluate proxy confounding.
r_{eff}	Cloud droplet effective radius from MODIS.
H	Cloud depth, calculated from MODIS retrievals using the adiabaticity assumption.
$\mathrm{COD_w}$	Cloud Optical Depth from MODIS, weighted by the grid-box cloud fraction (also from MODIS).
SW $\uparrow$	Reflected top-of-atmosphere SW radiation flux from CERES.
$\mathrm{RH_{BL}}$	Relative humidity from ERA5, averaged over pressure levels below the boundary layer height, as identified using the eponymous variable in ERA5.
$\mathrm{U_{700}}$	Eastward wind speed from ERA5, chosen at 700 hPa to capture mid-tropospheric large-scale transport by the Southern African Easterly Jet (AEJ-S, Adebiyi and Zuidema 2016; Pistone et al 2021).

2.2. Causal inference

The causal workflow includes 4 steps, summarized in Fig. 2: 1) assumption of a groundtruth causal graph, 2) computation of partial sensitivities from observations, with control variables dictated by the graph, 3) computation of total and mediated effects from the partial sensitivities, and 4) comparison to other causal graphs and computation of resulting biases.

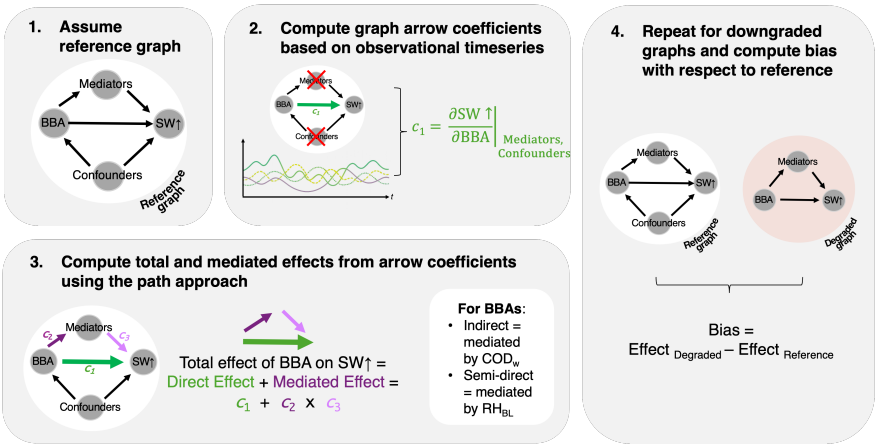

Figure 2. *Summary of the method, decomposed in 4 steps, from (1.) assumption of the reference causal graph to (4.) computation of biases in radiative effect estimates.*

Fig. 3 shows the 'Reference graph', describing the pathways by which BBAs affect the SW radiation budget of the SEA region (Step 1). Its structure is hypothesized from expert knowledge and is not obtained from data-driven techniques such as causal discovery (Runge et al 2023). Like in Fig. 1, the colors of the arrow identifiers a_n highlight the direct (ARI), indirect (ACI, cloud-mediated) and semi-direct (adjustments to ARI, humidity-mediated) pathways through which BBAs modify SW radiation at TOA.

Causal effect estimation is done with the Python Tigramite package[1] following Wright's linear path method (Runge et al 2015; Wright 1921). The estimation starts with the derivation of partial sensitivities, i.e., the values for each causal arrow in the graph, yielding the arrow colors in Fig. 3 (Step 2). Once the causal links have been estimated, causal effects between two variables that are not directly linked by an arrow are calculated by tracing the path(s) that link the two variables and multiplying the coefficients along these paths (Step 3). This is formalized in Eq. 2.1, which shows how the total shortwave radiative effect of BBAs is obtained with a linear multiplication of partial senstivities:

$$\mathrm{RE}_{\mathrm{BBA}\to\mathrm{SW}} = -\frac{\mathrm{dSW}\uparrow}{\mathrm{dBBA}} \times \Delta\mathrm{BBA}_{\mathrm{Forcing}}, \text{ where : } \underbrace{\frac{\mathrm{dSW}\uparrow}{\mathrm{dBBA}}}_{\text{Total effect}} = \left.\frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{BBA}}\right|_{\mathrm{direct}} + \left.\frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{BBA}}\right|_{\mathrm{indirect}} + \left.\frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{BBA}}\right|_{\mathrm{semi-direct}}$$

$$= \frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{BBA}} + \frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{COD_w}}\frac{\partial \mathrm{COD_w}}{\partial \mathrm{BBA}} + \frac{\partial \mathrm{SW}\uparrow}{\partial \mathrm{RH_{BL}}}\frac{\partial \mathrm{RH_{BL}}}{\partial \mathrm{BBA}} = a_1 + a_2a_3a_4 + \underbrace{a_5a_6}_{\substack{\text{non-cloud}\\ \text{semi-direct}}} + \underbrace{(a_5a_7 + a_5a_8\,(a_9 + a_{10}a_3))a_4}_{\substack{\text{cloud-mediated}\\ \text{semi-direct}}},$$

$$\text{and: } \Delta\mathrm{BBA}_{\mathrm{Forcing}} = \overline{\mathrm{BBA}_{\substack{25\%\ \mathrm{Smokiest}\\ \mathrm{days}}}} - \overline{\mathrm{BBA}_{\substack{25\%\ \mathrm{Least}\\ \mathrm{smoky\ days}}}} \text{ (see Fig. A.2)} \tag{2.1}$$

To estimate the impact of the main sources of statistical biases (Step 4), we repeat Steps 1 to 3 for 'degraded' variants of the graph, as described in Table 2 and Fig. A.1.

2.3. Practical considerations: linearity, spatial aggregation, uncertainty intervals

It can be questioned whether smoke-cloud-radiation interactions are linear (e.g., see Doherty et al 2022). Lohmann and Feichter 2001 found that linear additivity of the direct, indirect and semi-direct radiative

[1] https://github.com/jakobrunge/tigramite

Table 2. *Description of the causal graph variants considered in this study.*

Graph name	Arrows in Fig. 3	Description
Reference graph	All arrows in Fig. 3	Reference graph against which all the other confounded graphs are evaluated
AOD confounding	Same as reference graph, only the data source changes	AOD is used as a proxy for the BBA variable, introducing confounding as AOD is dependent upon RH_{BL} (Gryspeerdt et al 2016; Stier 2016)
Wind confounding	Reference graph minus the arrows pointing out of U_{700}	The westward AEJ-S, approximated by U_{700}, transports smoke, but also humidity, and influences subsidence and boundary layer circulations (Adebiyi and Zuidema 2016; Pistone et al 2021). Omitting U_{700} thus introduces confounding.
Regime confounding	Reference graph, but r_{eff} and H are omitted and replaced by a direct arrow from BBA to COD_w	Regimes of clouds with different depths are not explicitly separated, causing confounding (Fons et al 2023).
Auto-dependency	Reference graph minus the autodependency links	For all variables X, $X(t+1\ \mathrm{day})$ is considered independent from $X(t)$
Lagged feedbacks	Reference graph minus the lagged links from COD_w to BBA and RH_{BL}	Lagged effects of clouds on aerosol and humidity fields are removed
Spatial aggregation	Same as 'Reference graph', but the data normalization and the graph coefficients are calculated using the 'multidata' Tigramite option and not separately in each $3^\circ \times 3^\circ$ gridbox	The results are spatially generalizable but do not take into account potential latent spatially varying meteorological confounders
Spatial aggregation + Longitude	Same as 'Spatial Aggregation' but 'Longitude' is added as a graph node which confounds all the other variables	A very simple form of Earth Embedding aimed at reducing confounding by spatial aggregation. This acknowledges that different meteorogology-cloud-aerosol regimes exist, mostly along the longitude axis.

effects of BBAs was a valid assumption on a global scale. To test this with our regional data, we do a multiple linear regression of SW ↑ with respect to its parents variables in the reference graph, namely, BBA, COD_w, RH_{BL} and SW $\uparrow(t-1)$. On average for the region, the parent variables explain 76 % of the variability in the TOA SW. We thus consider the linear approximation to be reasonable.

Because aggregation over large regions can induce spurious statistical associations due to spatially covarying meteorological influences (Grandey and Stier 2010), we do not carry out the causal effect estimation of the large SEA region at once, but rather in sub-domains (Loeb and Schuster 2008) of size $3^\circ \times 3^\circ$, each containing 9 ($1^\circ \times 1^\circ$) grid points. The data are normalized and de-seasonalized by removing the mean seasonal cycle and dividing by the standard deviation of the seasonal cycle in each sub-domain. This means that the structure of the causal graph is spatially generalizable over the SEA region, but the causal effects are not. With the 'Spatial Aggregation' graph of Table 2, we bypass this spatial segmentation and pool all sub-domains using multidata causal inference to compute spatially uniform causal effects. 'Spatial Aggregation + Longitude' is an attempt at correcting the resulting aggregation bias while retaining spatial generalization of the results.

The uncertainties for the radiative effect estimates are computed with standard error propagation of the 68.2 % ($\pm\sigma_{\mathrm{dev}}$) bootstrap error intervals (with 100 bootstrap members) around the estimates of the

causal links. Maps showing causal effects and radiative effects feature stippling where the effects are not statistically different from 0, i.e., where the confidence interval includes 0.

3. Results and discussion

3.1. Causal graph validation

Fig. 3 shows the computed values of every causal arrow featured in the reference causal graph (Step 2 in Fig. 2). The causal links are resolved spatially on a 3° × 3° grid (Fig. A.3) but a domain-wide mean is shown here. The analysis of these values constitutes a physical validation of the method as the signs of the effects correspond to the direction of the expected physical mechanisms (Table 3).

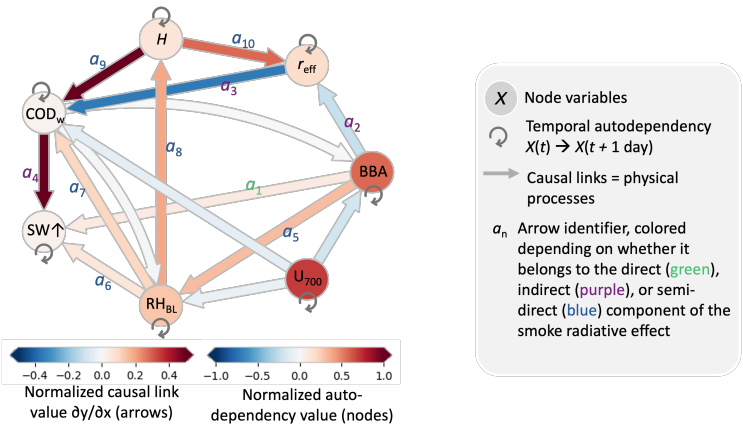

Figure 3. *Quantified causal links for the reference graph. Arrow colors correspond to the normalized magnitude of causal links between two variables, for contemporaneous effects (straight arrows) and lagged effects (curved arrows), while node colors correspond to magnitude of autodependency links.*

3.2. Total shortwave effect of biomass burning aerosols

Fig. 4 shows the total SW radiative effect of BBAs, as well as the fractions of that effect that are direct (non-mediated), indirect (cloud-mediated), and semi-direct (RH_{BL}-mediated), as calculated using the path approach applied to the causal links computed just above (Step 3 in Fig. 2). The smoke emitted over the SEA stratocumulus region during the biomass burning season causes a net regional cooling of $\approx -2.5\,\mathrm{W\,m^{-2}}$. The direct, indirect and semi-direct pathways contribute almost equally to this cooling.

The warming/cooling dipole in total BBA radiative effect from east to west is mostly attributable to the direct radiative effect. This is due to the presence of dark absorbing aerosols overlying a dense bright stratocumulus deck in the east, leading to a local warming of up to $\approx 1\,\mathrm{W\,m^{-2}}$. In the west, cloud cover is sparser, and organic aerosols have photochemically 'whitened' and become more efficient at scattering during atmospheric transport (e.g., Carter et al 2021), leading to local cooling effects of more than $-2\,\mathrm{Wm^{-2}}$. Similar patterns have been found in reanalysis and model data, but model results are highly dependent on their choice of physical parameters and forcing methodologies (Giuffrida et al 2025; Mallet et al 2020; Myhre et al 2013; Zuidema et al 2016). Other observational studies (Doherty et al 2022; Kacenelenbogen et al 2019) have found stronger positive direct radiative effects of BBAs over the eastern SEA. However, the sampling of high-smoke events in these studies could have resulted

Table 3. *Physical interpretation of the reference graph's causal links (Fig. 3).*

Arrow in the Reference Graph	Computed Sign	Physical Interpretation
BBA → SW↑	+	Direct aerosol effect (some absorption by BC in the east of the domain but mostly scattering by OC in the west)
COD_w → SW↑	+	Strong albedo of clouds over ocean
RH_{BL} → SW↑	+	The only effect that is not as expected. Could be due to a confounding influence from temperature (e.g., less efficient water vapor absorption of SW at colder temperature, Wang et al 2022), or an imperfect separability of water vapor and cloud effects due to method limitations.
BBA → r_{eff}	-	Aged hygroscopic BBAs can serve as CCN, leading to droplet size reduction at a constant liquid water content (Gupta et al 2021)
BBA → RH_{BL}	+	When above the BL, BBA can cause increased LTS, less entrainment and a moister BL (Koch and Del Genio 2010; Wilcox 2010)
H → r_{eff}	+	Adiabatic cloud assumption for thin low-level clouds (Lohmann et al 2016a)
RH_{BL} → H	+	Cloud deepening in moist BL
r_{eff} → COD_w	-	Twomey effect and potential positive cloud fraction adjustments (Gryspeerdt et al 2016; Twomey 1977)
H → COD_w	-	Depth-dependence of COD (by definition)
RH_{BL} → COD_w	+	RH-dependence of cloud formation (Köhler 1921)
U_{700} → BBA	-	Westward smoke transport by the AEJ-S (Adebiyi and Zuidema 2016)
U_{700} → COD_w	-	Secondary circulations induced by the AEJ-S (Adebiyi and Zuidema 2016)
U_{700} → RH_{BL}	-	Humidity advection (Baró Pérez et al 2024; Pistone et al 2021) and secondary circulations induced by the AEJ-S (Adebiyi and Zuidema 2016)
COD_w → BBA (lag 1)	-	Wet scavenging of below-cloud smoke by precipitation
COD_w → RH_{BL} (lag 1)	+/-	A superposition of various effects: precipitation, surface cooling, and turbulent mixing induced by cloud top cooling

in selection bias, whereas our estimates are representative of the whole duration of the biomass burning season in the SEA.

The indirect radiative effect is negative in the western part of the domain but slightly positive at the coast. In the west, this is due to aged and hygroscopic smoke being entrained into the boundary layer and acting as CCN (Gupta et al 2021), making clouds brighter and triggering cloud adjustments, like increased cloud cover and lifetime. In the east, freshly emitted BBAs are hydrophobic and cannot act as CCN (Redemann et al 2021). If hydrophobic smoke is entrained into the boundary layer, it competes with hydrophylic aerosols for water vapor, resulting in fewer and less bright cloud droplets (Ching et al 2016). However, it is also possible that thick smoke layers overlying clouds bias the retrievals of cloud properties (Alfaro-Contreras et al 2014). Modeling studies have also found negative indirect smoke radiative effects of the same magnitude, although some studies have a simplistic parameterization of aerosol-cloud interactions (Giuffrida et al 2025; Lohmann et al 2000; Rap et al 2013).

While models have been shown to disagree on the sign and magnitude of the semi-direct effect (Allen et al 2019; Mallet et al 2020; Sakaeda et al 2011), we detect uniform cooling, primarily driven by the cloud-mediated semi-direct effect (-0.7 W m^{-2}) with a smaller contribution from the non-cloud semi-direct effect (-0.2 W m^{-2}).

Our observational results highlight that the total smoke radiative effect results from almost equally cooling contributions of the direct, indirect and semi-direct pathways. This finding can be used to

constrain climate models that have previously found that the indirect effect dominates the total cooling (e.g., Lohmann and Feichter 2001; Lu et al 2018).

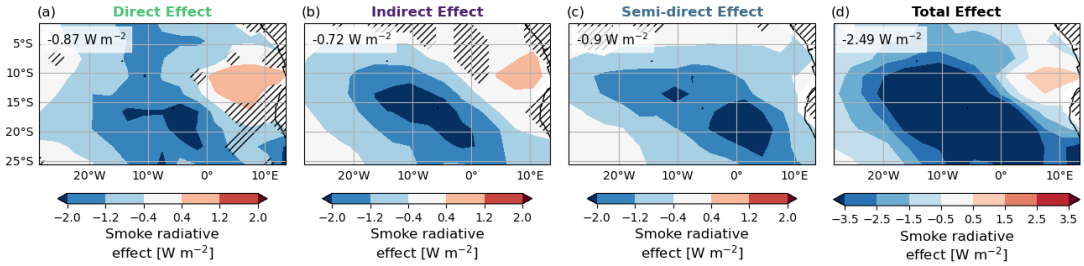

Figure 4. *BBA radiative effect per pathway:* ***(a)*** *direct,* ***(b)*** *indirect,* ***(c)*** *semi-direct and* ***(d)*** *resulting total (note the different colorscale). The region-average BBA shortwave radiative effect is displayed in the upper left corner. Stippling indicates where the bootstrap confidence intervals include 0.*

3.3. Sources of confounding

In Step 4 of the analysis, the reference graph is modified (Table 2) to single out sources of confounding bias.

Spatial aggregation results in the largest confounding bias with a -51 % cooling bias compared to the reference graph. This is mostly due to the longitudinal gradient of meteorological regimes present in the SEA, with very bright clouds and a thick smoke layer in the east, but more dispersed clouds and less aerosols in the pristine western part of the domain, creating a spurious positive correlation between aerosols, cloud brightness and reflected radiation. However, when simply adding longitude as a confounder node in the causal graph, the large bias almost vanishes (from -51% to +5%). This simple yet powerful change allows for spatial generalization while keeping bias at a minimum.

Diamond et al 2018 pointed out the time-dependence of smoke-cloud interactions over the SEA. Here we detect that neglecting auto-dependency of meteorological fields, and especially of RH_{BL}, U_{700} and BBA (see node colors in Fig. 3) leads to a 19% overestimation of the cooling influence of BBAs. It would be interesting to repeat the present analyses with a Lagrangian perspective to identify the exact processes responsible for this behavior.

Omitting r_{eff} as a mediator of the indirect radiative effect of BBAs leads to an underestimation of the cooling influence of BBAs (by 14%). This is consistent with the findings of Fons et al 2023, and is due to an overestimation of cloud optical depth decreases when aggregating data over different cloud regimes.

The AOD cooling bias (-11%) is due to the humidity-dependence of the AOD proxy, resulting in a spuriously enhanced correlation between RH and AOD (e.g., Gryspeerdt et al 2016; Stier 2016).

Neglecting confounding from large-scale winds leads to a small overestimation of the cooling influence of BBAs. This is coherent with Baró Pérez et al 2024, who found a radiative bias due to moisture accompanying the smoke plume, although their cooling bias was much stronger than ours.

Importantly, while different sources of confounding might compensate each other to some degree, ignoring any of them would result in a loss of robustness as there is a lack of generalization power when confounding is present.

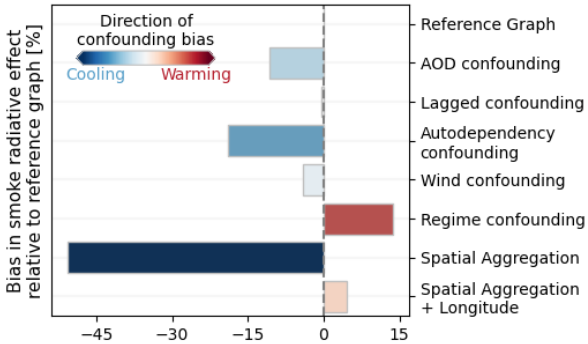

Figure 5. *Biases in smoke radiative effect caused by different sources of confounding.*

4. Conclusions and perspectives

We applied causal inference to estimate the observed radiative effect of smoke aerosols over the South-East Atlantic. We find that BBAs cause a total radiative effect of -2.5 W m^{-2} with approximately equal contributions from the direct, indirect and semi-direct effects. Contrary to many modeling studies that find strong warming direct effects of BBAs, we find moderate warming (up to 1 W m^{-2}) that is confined in the east of the SEA basin, where freshly emitted smoke overlies a bright stratocumulus decks. Using variants of our causal graph, we identify the main sources of confounding: aggregation across space and cloud regimes, temporal autocorrelation, using AOD as a proxy for BBA aerosols and large-scale dynamical influences from the Southern-African Easterly Jet. This graph comparison highlights the key physical variables that need to be taken into account for smoke-cloud-radiation studies and for which measurement errors need to be reduced in future satellite developments or field campaigns. Some limitations of this study deserve to be investigated in future work, for instance, the ground truth assumption for the reference causal graph, the linear assumption, the Eulerian nature of the analyses, or satellite retrieval biases. Despite these limitations, causal inference proves to be a helpful tool to decompose the complex pathways through which biomass burning aerosols modify the radiative budget of the South-East Atlantic under complex confounding influences from meteorology, providing bias-free observational constraints that can guide process-based evaluations of climate models. Climate models with better smoke parameterizations will eventually be useful to evaluate the effects of wildfire smoke and pollution in the anthropogenic era. At the global scale, this could help increase confidence in future temperature projections. At the regional scale, this could help evaluate smoke- and pollution-driven changes in the hydrological cycle and thus inform adaptation measures related to droughts or flooding.

Funding Statement. This work was supported by the European Union's Horizon 2020 research and innovation program under Marie Sklodowska-Curie grant agreement No. 860100 (iMIRACLI). ILM research was supported by NOAA cooperative agreement NA22OAR4320151, for the Cooperative Institute for Earth System Research and Data Science (CIESRDS). We acknowledge the International Space Science Institute (ISSI) in Bern, Switzerland who facilitated discussions of these concepts through ISSI International Team Project 23-576.

Competing Interests. The authors declare no competing interest.

Code and Data Availability Statement. The source data used in this study can be downloaded from: MODIS (https://ladsweb.modaps.eosdis.nasa.gov/archive/allData/61/MOD08_D3/), CERES (https://ceres.larc.nasa.gov/data/), ERA5 (https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels), CAMS (https://ads.atmosphere.copernicus.eu/datasets/cams-global-reanalysis-eac4). The analysis and plotting code is available on Zenodo (Fons et al 2026b). Intermediate data produced by and used by the code are available on Zenodo as well (Fons et al 2026a).

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. Conceptualization: E.F.; U.L.; D.N. Methodology: E.F.; I.L.M.; T.B. Data curation: E.F. Data visualisation: E.F.. Writing original draft: E.F. All authors approved the final submitted draft.

A. Appendix

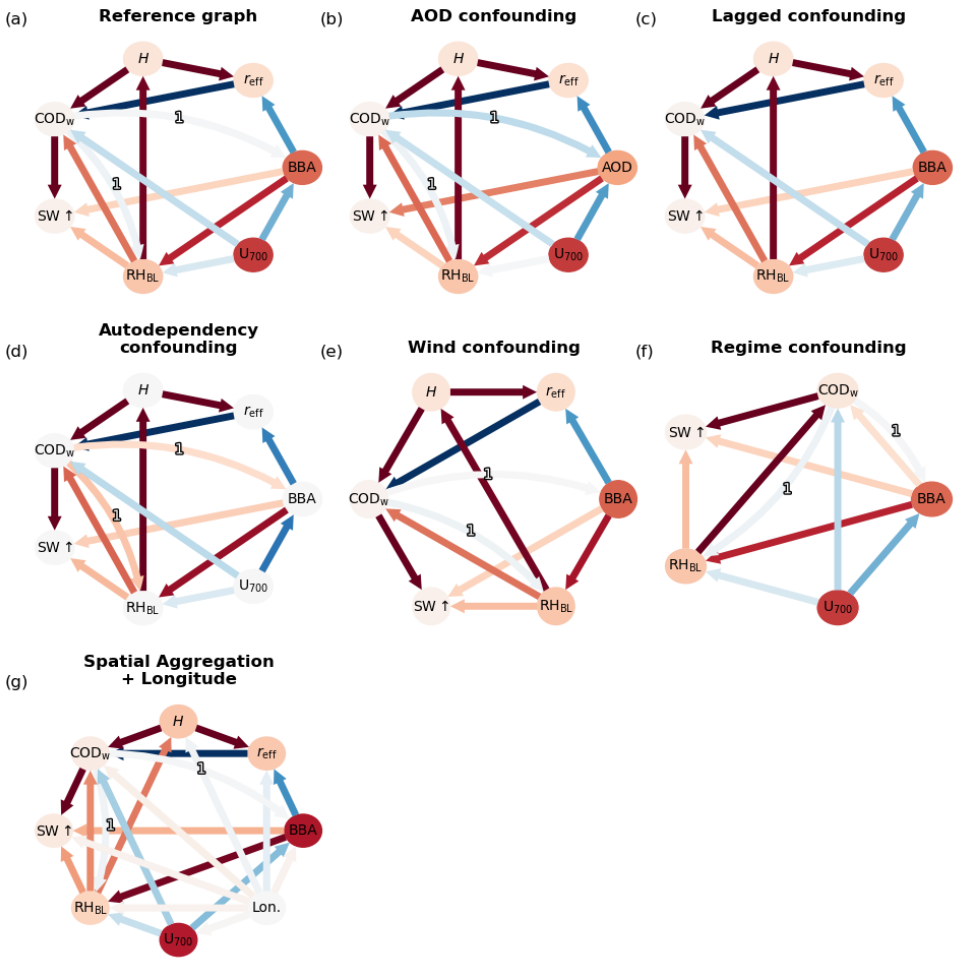

Figure A.1. *All causal graph variants evaluated in the study, as described in Table 2. The arrow colorscale is bounded by -0.2 and 0.2, the node colorscale by -1 and 1.*

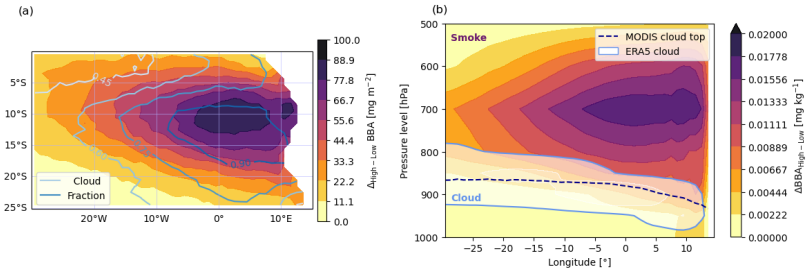

Figure A.2. *Smoke and cloud fields over the SEA during the biomass burning season. (a) Filled contours show* $\Delta\mathrm{BBA}_{\mathrm{Forcing}}$ *(defined in Eq. 2.1) with overlain contours of MODIS cloud fraction. (b) Same but as a vertical cross-section, averaged over latitude. Overlain are the mean low-level cloud layer computed from ERA5 liquid water content (*$\geq 10\,\mathrm{g\,kg^{-1}}$*), and the mean cloud top pressure from MODIS.*

References

Adebiyi AA and Zuidema P (2016) The Role of the Southern African Easterly Jet in Modifying the Southeast Atlantic Aerosol and Cloud Environments. *Quarterly Journal of the Royal Meteorological Society* **142**(697), 1574–1589. doi: 10.1002/qj.2765.

Alfaro-Contreras R, Zhang J, Campbell JR, Holz RE and Reid JS (2014) Evaluating the Impact of Aerosol Particles above Cloud on Cloud Optical Depth Retrievals from MODIS. *Journal of Geophysical Research: Atmospheres* **119**(9), 5410–5423. doi: 10.1002/2013JD021270.

Allen RJ, Amiri-Farahani A, Lamarque JF, Smith C, Shindell D, Hassan T and Chung CE (2019) Observationally Constrained Aerosol–Cloud Semi-Direct Effects. *npj Climate and Atmospheric Science* **2**(1), 1–12. doi: 10.1038/s41612-019-0073-9.

Baró Pérez A, Diamond MS, Bender FAM, Devasthale A, Schwarz M, Savre J, Tonttila J, Kokkola H, Lee H, Painemal D and Ekman AML (2024) Comparing the simulated influence of biomass burning plumes on low-level clouds over the southeastern Atlantic under varying smoke conditions. *Atmospheric Chemistry and Physics* **24**(8), 4591–4610. doi: 10.5194/acp-24-4591-2024. https://acp.copernicus.org/articles/24/4591/2024/.

Bellouin N, Quaas J, Gryspeerdt E, Kinne S, Stier P, Watson-Parris D, Boucher O, Carslaw KS, Christensen M, Daniau AL, Dufresne JL, Feingold G, Fiedler S, Forster P, Gettelman A, Haywood JM, Lohmann U, Malavelle F, Mauritsen T, McCoy DT, Myhre G, Mülmenstädt J, Neubauer D, Possner A, Rugenstein M, Sato Y, Schulz M, Schwartz SE, Sourdeval O, Storelvmo T, Toll V, Winker D and Stevens B (2020) Bounding Global Aerosol Radiative Forcing of Climate Change. *Reviews of Geophysics* **58**(1), e2019RG000660. doi: 10.1029/2019RG000660.

Brenguier JL, Pawlowska H, Schüller L, Preusker R, Fischer J and Fouquart Y (2000) Radiative Properties of Boundary Layer Clouds: Droplet Effective Radius versus Number Concentration. *Journal of the Atmospheric Sciences* **57**(6), 803–821. doi: 10.1175/1520-0469(2000)057<0803:RPOBLC>2.0.CO;2.

Carter TS, Heald CL, Cappa CD, Kroll JH, Campos TL, Coe H, Cotterell MI, Davies NW, Farmer DK, Fox C, Garofalo LA, Hu L, Langridge JM, Levin EJT, Murphy SM, Pokhrel RP, Shen Y, Szpek K, Taylor JW and Wu H (2021) Investigating Carbonaceous Aerosol and Its Absorption Properties From Fires in the Western United States (WE-CAN) and Southern Africa (ORACLES and CLARIFY). *Journal of Geophysical Research: Atmospheres* **126**(15), e2021JD034984. doi: 10.1029/2021JD034984.

Ching J, Riemer N and West M (2016) Black carbon mixing state impacts on cloud microphysical properties: Effects of aerosol plume and environmental conditions. en. *Journal of Geophysical Research: Atmospheres*, 23 May **121**(10), _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/2016JD024851, 5990–6013. ISSN: 2169-8996. doi: 10.1002/2016JD024851. https://onlinelibrary.wiley.com/doi/abs/10.1002/2016JD024851 (accessed 23 May 2024).

Diamond M, Dobracki A, Freitag S, Small-Griswold J, Heikkila A, Howell S, Kacarab M, Podolske J, Saide P and Wood R (2018) Time-Dependent Entrainment of Smoke Presents an Observational Challenge for Assessing Aerosol–Cloud Interactions over the Southeast Atlantic Ocean. *Atmospheric Chemistry and Physics* **18**(19), 14623–14636. doi: 10.5194/acp-18-14623-2018.

Diamond M, Saide P, Zuidema P, Ackerman A, Doherty S, Fridlind A, Gordon H, Howes C, Kazil J, Yamaguchi T, Zhang J, Feingold G and Wood R (2022) Cloud Adjustments from Large-Scale Smoke–Circulation Interactions Strongly Modulate the Southeastern Atlantic Stratocumulus-to-Cumulus Transition. *Atmospheric Chemistry and Physics* **22**(18), 12113–12151. doi: 10.5194/acp-22-12113-2022.

Doelling DR, Loeb NG, Keyes DF, Nordeen ML, Morstad D, Nguyen C, Wielicki BA, Young DF and Sun M (2013) Geostationary Enhanced Temporal Interpolation for CERES Flux Products. EN. *Journal of Atmospheric and Oceanic Technology*, 26 March **30**(6), 1072–1090. ISSN: 0739-0572, 1520-0426. doi: 10.1175/JTECH-D-12-00136.1 (accessed 26 March 2026).

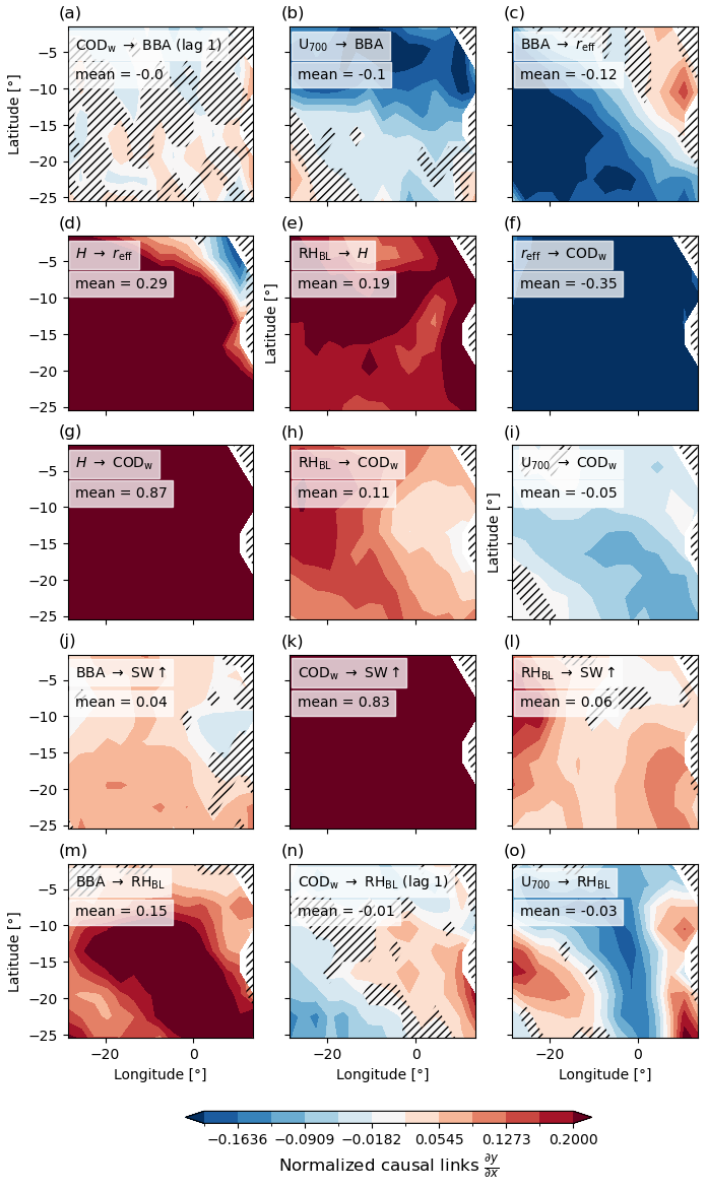

Figure A.3. *Spatial distribution of normalized causal links for the reference graph. The subplots show the sign and magnitude of causal effects over the SEA region, for each causal arrows in the graph (Fig. 3), as indicated in the upper left corner of each subplot..*

Doelling DR, Sun M, Nguyen LT, Nordeen ML, Haney CO, Keyes DF and Mlynczak PE (2016) Advances in Geostationary-Derived Longwave Fluxes for the CERES Synoptic (SYN1deg) Product. EN. *Journal of Atmospheric and Oceanic Technology*, 26 March **33**(3), 503–521. ISSN: 0739-0572, 1520-0426. doi: 10.1175/JTECH-D-15-0147.1 (accessed 26 March 2026).

Doherty SJ, Saide PE, Zuidema P, Shinozuka Y, Ferrada GA, Gordon H, Mallet M, Meyer K, Painemal D, Howell SG, Freitag S, Dobracki A, Podolske JR, Burton SP, Ferrare RA, Howes C, Nabat P, Carmichael GR, da Silva A, Pistone K, Chang I, Gao L, Wood R and Redemann J (2022) Modeled and Observed Properties Related to the Direct Aerosol Radiative Effect of Biomass Burning Aerosol over the Southeastern Atlantic. *Atmospheric Chemistry and Physics* **22**(1), 1–46. doi: 10.5194/acp-22-1-2022.

Fons E, McCoy IL, Beucler T, Neubauer D and Lohmann U (2026a) *Data for Dissipating the correlation smokescreen: Causal decomposition of the radiative effects of biomass burning aerosols over the South-East Atlantic*. Climate Informatics Conference 2026, Lausanne. Zenodo. Zenodo. doi: 10.5281/zenodo.19236234.

Fons E, McCoy IL, Beucler T, Neubauer D and Lohmann U (2026b) Software for Dissipating the correlation smokescreen: Causal decomposition of the radiative effects of biomass burning aerosols over the South-East Atlantic. Climate Informatics Conference 2026, Lausanne. Zenodo. doi: 10.5281/zenodo.19222372.

Fons E, Runge J, Neubauer D and Lohmann U (2023) Stratocumulus Adjustments to Aerosol Perturbations Disentangled with a Causal Approach. *npj Climate and Atmospheric Science* **6**(1). doi: 10.1038/s41612-023-00452-w.

Ghan SJ (2013) Technical Note: Estimating Aerosol Effects on Cloud Radiative Forcing. *Atmospheric Chemistry and Physics* **13**(19), 9971–9974. doi: 10.5194/acp-13-9971-2013.

Giuffrida E, Johnson K, Tatro T, Zuidema P and Gordon H (2025) Biomass burning aerosol radiative effects in the Southeast Atlantic depend strongly on meteorological forcing method. *Atmospheric Chemistry and Physics* **25**(21), 14879–14907. doi: 10.5194/acp-25-14879-2025. https://acp.copernicus.org/articles/25/14879/2025/.

Grandey BS and Stier P (2010) A Critical Look at Spatial Scale Choices in Satellite-Based Aerosol Indirect Effect Studies. *Atmospheric Chemistry and Physics* **10**(23), 11459–11470. doi: 10.5194/acp-10-11459-2010.

Gryspeerdt E, Quaas J and Bellouin N (2016) Constraining the Aerosol Influence on Cloud Fraction. *Journal of Geophysical Research: Atmospheres* **121**(7), 3566–3583. doi: 10.1002/2015JD023744.

Gupta S, McFarquhar GM, O'Brien JR, Delene DJ, Poellot MR, Dobracki A, Podolske JR, Redemann J, LeBlanc SE, Segal-Rozenhaimer M and Pistone K (2021) Impact of the Variability in Vertical Separation between Biomass Burning Aerosols and Marine Stratocumulus on Cloud Microphysical Properties over the Southeast Atlantic. *Atmospheric Chemistry and Physics* **21**(6), 4615–4635. doi: 10.5194/acp-21-4615-2021.

Hersbach H, Bell B, Berrisford P, Biavati G, Horányi A, Muñoz Sabater J, Nicolas J, Peubey C, Radu R, Rozum I, Schepers D, Simmons A, Soci C, Dee D and Thébaut JN (2018a) ERA5 Hourly Data on Pressure Levels from 1959 to Present. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*, 27 February. doi: 10.24381/cds.bd0915c6 (accessed 27 February 2023).

Hersbach H, Bell B, Berrisford P, Biavati G, Horányi A, Muñoz Sabater J, Nicolas J, Peubey C, Radu R, Rozum I, Schepers D, Simmons A, Soci C, Dee D and Thébaut JN (2018b) ERA5 Hourly Data on Single Levels from 1959 to Present. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*, 27 February. doi: 10.24381/cds.adbb2d47 (accessed 27 February 2023).

Imbens G and Rubin D (2015) *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge: Cambridge University Press. doi: 10.1017/CBO9781139025751.

Inness A, Ades M, Agustí-Panareda A, Barré J, Benedictow A, Blechschmidt AM, Dominguez JJ, Engelen R, Eskes H, Flemming J, Huijnen V, Jones L, Kipling Z, Massart S, Parrington M, Peuch VH, Razinger M, Remy S, Schulz M and Suttie M (2019) The CAMS reanalysis of atmospheric composition. English. *Atmospheric Chemistry and Physics*, 2 February **19**(6), 3515–3556. ISSN: 1680-7316. doi: 10.5194/acp-19-3515-2019 (accessed 2 February 2024).

Kacenelenbogen MS, Vaughan MA, Redemann J, Young SA, Liu Z, Hu Y, Omar AH, LeBlanc S, Shinozuka Y, Livingston J, Zhang Q and Powell KA (2019) Estimations of Global Shortwave Direct Aerosol Radiative Effects above Opaque Water Clouds Using a Combination of A-Train Satellite Sensors. *Atmospheric Chemistry and Physics* **19**(7), 4933–4962. doi: 10.5194/acp-19-4933-2019.

Koch D and Del Genio AD (2010) Black Carbon Semi-Direct Effects on Cloud Cover: Review and Synthesis. *Atmospheric Chemistry and Physics* **10**(16), 7685–7696. doi: 10.5194/acp-10-7685-2010.

Köhler H (1921) *Zur Kondensation des Wasserdampfes in der Atmosphäre*. I kommission hos Cammermeyers boghandel Kristiania.

Loeb NG and Schuster GL (2008) An Observational Study of the Relationship between Cloud, Aerosol and Meteorology in Broken Low-Level Cloud Conditions. *Journal of Geophysical Research: Atmospheres*, 26 November **113**(D14). doi: 10.1029/2007JD009763 (accessed 26 November 2020).

Lohmann U and Feichter J (2001) Can the Direct and Semi-Direct Aerosol Effect Compete with the Indirect Effect on a Global Scale? *Geophysical Research Letters* **28**(1), 159–161. doi: 10.1029/2000GL012051.

Lohmann U, Feichter J, Penner J and Leaitch R (2000) Indirect Effect of Sulfate and Carbonaceous Aerosols: A Mechanistic Treatment. *Journal of Geophysical Research: Atmospheres* **105**(D10), 12193–12206. doi: 10.1029/1999JD901199.

Lohmann U, Lüönd F and Mahrt F (2016a) Global Energy Budget. In *An Introduction to Clouds: From the Microscale to Climate*. Cambridge University Press, 323–334.

Lohmann U, Lüönd F and Mahrt F (2016b) Microphysical processes in warm clouds. In *An Introduction to Clouds: From the Microscale to Climate*. Cambridge University Press, 186–217.

Lu Z, Liu X, Zhang Z, Zhao C, Meyer K, Rajapakshe C, Wu C, Yang Z and Penner JE (2018) Biomass Smoke from Southern Africa Can Significantly Enhance the Brightness of Stratocumulus over the Southeastern Atlantic Ocean. *Proceedings of the National Academy of Sciences* **115**(12), 2924–2929. doi: 10.1073/pnas.1713703115.

Mallet M, Solmon F, Nabat P, Elguindi N, Waquet F, Bouniol D, Sayer AM, Meyer K, Roehrig R, Michou M, Zuidema P, Flamant C, Redemann J and Formenti P (2020) Direct and Semi-Direct Radiative Forcing of Biomass-Burning

Aerosols over the Southeast Atlantic (SEA) and Its Sensitivity to Absorbing Properties: A Regional Climate Modeling Study. *Atmospheric Chemistry and Physics* **20**(21), 13191–13216. doi: 10.5194/acp-20-13191-2020.

Myhre G, Samset BH, Schulz M, Balkanski Y, Bauer S, Berntsen TK, Bian H, Bellouin N, Chin M, Diehl T, Easter RC, Feichter J, Ghan SJ, Hauglustaine D, Iversen T, Kinne S, Kirkevåg A, Lamarque JF, Lin G, Liu X, Lund MT, Luo G, Ma X, van Noije T, Penner JE, Rasch PJ, Ruiz A, Seland Ø, Skeie RB, Stier P, Takemura T, Tsigaridis K, Wang P, Wang Z, Xu L, Yu H, Yu F, Yoon JH, Zhang K, Zhang H and Zhou C (2013) Radiative Forcing of the Direct Aerosol Effect from AeroCom Phase II Simulations. *Atmospheric Chemistry and Physics* **13**(4), 1853–1877. doi: 10.5194/acp-13-1853-2013.

Parmesan C, Morecroft M, Trisurat Y, Adrian R, Anshari G, Arneth A, Gao Q, Gonzalez P, Harris R, Price J, Stevens N and Talukdarr G (2022) Terrestrial and Freshwater Ecosystems and Their Services. In *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)] doi: doi:10.1017/9781009325844.004.

Pearl J (2009) *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge: Cambridge University Press. doi: 10.1017/CBO9780511803161.

Pistone K, Zuidema P, Wood R, Diamond M, da Silva AM, Ferrada G, Saide PE, Ueyama R, Ryoo JM, Pfister L, Podolske J, Noone D, Bennett R, Stith E, Carmichael G, Redemann J, Flynn C, LeBlanc S, Segal-Rozenhaimer M and Shinozuka Y (2021) Exploring the Elevated Water Vapor Signal Associated with the Free Tropospheric Biomass Burning Plume over the Southeast Atlantic Ocean. *Atmospheric Chemistry and Physics* **21**(12), 9643–9668. doi: 10.5194/acp-21-9643-2021.

Platnick S, King M and Hubanks P (2015) *MODIS Atmosphere L3 Daily Product. NASA MODIS Adaptive Processing System, Goddard Space Flight Center, USA*. doi: http://dx.doi.org/10.5067/MODIS/MOD08_D3.006.

Quaas J, Boucher O and Lohmann U (2006) Constraining the Total Aerosol Indirect Effect in the LMDZ and ECHAM4 GCMs Using MODIS Satellite Data. *Atmospheric Chemistry and Physics* **6**(4), 947–955. doi: 10.5194/acp-6-947-2006.

Rap A, Scott CE, Spracklen DV, Bellouin N, Forster PM, Carslaw KS, Schmidt A and Mann G (2013) Natural Aerosol Direct and Indirect Radiative Effects. *Geophysical Research Letters* **40**(12), 3297–3301. doi: 10.1002/grl.50441.

Redemann J, Wood R, Zuidema P, Doherty SJ, Luna B, LeBlanc SE, Diamond MS, Shinozuka Y, Chang IY, Ueyama R, Pfister L, Ryoo JM, Dobracki AN, da Silva AM, Longo KM, Kacenelenbogen MS, Flynn CJ, Pistone K, Knox NM, Piketh SJ, Haywood JM, Formenti P, Mallet M, Stier P, Ackerman AS, Bauer SE, Fridlind AM, Carmichael GR, Saide PE, Ferrada GA, Howell SG, Freitag S, Cairns B, Holben BN, Knobelspiesse KD, Tanelli S, L'Ecuyer TS, Dzambo AM, Sy OO, McFarquhar GM, Poellot MR, Gupta S, O'Brien JR, Nenes A, Kacarab M, Wong JPS, Small-Griswold JD, Thornhill KL, Noone D, Podolske JR, Schmidt KS, Pilewskie P, Chen H, Cochrane SP, Sedlacek AJ, Lang TJ, Stith E, Segal-Rozenhaimer M, Ferrare RA, Burton SP, Hostetler CA, Diner DJ, Seidel FC, Platnick SE, Myers JS, Meyer KG, Spangenberg DA, Maring H and Gao L (2021) An Overview of the ORACLES (ObseRvations of Aerosols above CLouds and Their intEractionS) Project: Aerosol–Cloud–Radiation Interactions in the Southeast Atlantic Basin. *Atmospheric Chemistry and Physics* **21**(3), 1507–1563. doi: 10.5194/acp-21-1507-2021.

Reid C, Brauer M, Johnston F, Jerrett M, Balmes J and Elliott C (2016) Critical Review of Health Impacts of Wildfire Smoke Exposure. *Environmental Health Perspectives* **124**(9), 1334–1343. doi: 10.1289/ehp.1409277.

Reid JS, Eck TF, Christopher SA, Koppmann R, Dubovik O, Eleuterio DP, Holben BN, Reid EA and Zhang J (2005) A Review of Biomass Burning Emissions Part III: Intensive Optical Properties of Biomass Burning Particles. *Atmospheric Chemistry and Physics* **5**(3), 827–849. doi: 10.5194/acp-5-827-2005.

Runge J, Gerhardus A, Varando G, Eyring V and Camps-Valls G (2023) Causal Inference for Time Series. *Nature Reviews Earth & Environment*, 1–19. doi: 10.1038/s43017-023-00431-y.

Runge J, Petoukhov V, Donges JF, Hlinka J, Jajcay N, Vejmelka M, Hartman D, Marwan N, Paluš M and Kurths J (2015) Identifying Causal Gateways and Mediators in Complex Spatio-Temporal Systems. *Nature Communications* **6**(1), 8502. doi: 10.1038/ncomms9502.

Sakaeda N, Wood R and Rasch P (2011) Direct and Semidirect Aerosol Effects of Southern African Biomass Burning Aerosol. *J. Geophys. Res* **116**. doi: 10.1029/2010JD015540.

Stier P, Schutgens NaJ, Bellouin N, Bian H, Boucher O, Chin M, Ghan S, Huneeus N, Kinne S, Lin G, Ma X, Myhre G, Penner JE, Randles CA, Samset B, Schulz M, Takemura T, Yu F, Yu H and Zhou C (2013) Host Model Uncertainties in Aerosol Radiative Forcing Estimates: Results from the AeroCom Prescribed Intercomparison Study. *Atmospheric Chemistry and Physics* **13**(6), 3245–3270. doi: 10.5194/acp-13-3245-2013.

Stier P (2016) Limitations of Passive Remote Sensing to Constrain Global Cloud Condensation Nuclei. *Atmospheric Chemistry and Physics* **16**(10), 6595–6607. doi: 10.5194/acp-16-6595-2016.

Stone RS, Augustine JA, Dutton EG, O'Neill NT and Saha A (2011) Empirical determinations of the longwave and shortwave radiative forcing efficiencies of wildfire smoke. en. *Journal of Geophysical Research: Atmospheres*, 8 January **116**(D12). ISSN: 2156-2202. doi: 10.1029/2010JD015471. https://onlinelibrary.wiley.com/doi/abs/10.1029/2010JD015471 (accessed 8 January 2026).

Szopa S, Naik V, Adhikary B, Artaxo P, Berntsen T, Collins W, Fuzzi S, Gallardo L, Kiendler-Scharr A, Klimont Z, Liao H, Unger N and Zanis P (2021) Short-lived Climate Forcers. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, United

Kingdom and New York, NY, USA: [Masson-Delmotte, V., P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu and B. Zhou (eds.)] Cambridge University Press, 817–922. doi: doi:10.1017/9781009157896.008.

Twomey S (1977) The Influence of Pollution on the Shortwave Albedo of Clouds. *Journal of the Atmospheric Sciences* **34**(7), 1149–1152. doi: 10.1175/1520-0469(1977)034<1149:TIOPOT>2.0.CO;2.

van Marle MJE, Kloster S, Magi BI, Marlon JR, Daniau AL, Field RD, Arneth A, Forrest M, Hantson S, Kehrwald NM, Knorr W, Lasslop G, Li F, Mangeon S, Yue C, Kaiser JW and van der Werf GR (2017) Historic Global Biomass Burning Emissions for CMIP6 (BB4CMIP) Based on Merging Satellite Observations with Proxies and Fire Models (1750–2015). *Geoscientific Model Development* **10**(9), 3329–3357. doi: 10.5194/gmd-10-3329-2017.

Wang ZC, Yin B, Min Q and Zhu L (2022) Temperature-Dependence of the near-UV Absorption of Water Vapor in the 290–350nm Range. *Journal of Quantitative Spectroscopy and Radiative Transfer* **286**, 108204. doi: 10.1016/j.jqsrt.2022.108204.

Wilcox EM (2010) Stratocumulus Cloud Thickening beneath Layers of Absorbing Smoke Aerosol. *Atmospheric Chemistry and Physics* **10**(23), 11769–11777. doi: 10.5194/acp-10-11769-2010.

Wright S (1921) Correlation and Causation. *Journal of Agricultural Research* **20**, 557–585.

Zhong Q, Schutgens N, Werf GR van der, Noije T van, Bauer SE, Tsigaridis K, Mielonen T, Checa-Garcia R, Neubauer D, Kipling Z, Kirkevåg A, Olivié DJL, Kokkola H, Matsui H, Ginoux P, Takemura T, Le Sager P, Rémy S, Bian H and Chin M (2022) Using modelled relationships and satellite observations to attribute modelled aerosol biases over biomass burning regions. *Nature Communications* **13**(1), 5914. doi: 10.1038/s41467-022-33680-4.

Zhuang J, Dussin R, Huard D, Bourgault P, Banihirwe A, Raynaud S, Malevich B, Schupfner M, Levang S, Gauthier C et al (2023) pangeo-data/xESMF: v0. 8.2. *Zenodo* **8356796**. doi: 10.5281/zenodo.4294774.

Zuidema P, Redemann J, Haywood J, Wood R, Piketh S, Hipondoka M and Formenti P (2016) Smoke and Clouds above the Southeast Atlantic: Upcoming Field Campaigns Probe Absorbing Aerosol's Impact on Climate. *Bulletin of the American Meteorological Society* **97**(7), 1131–1135. doi: 10.1175/BAMS-D-15-00082.1.